14 Year-end review and consolidation of Seconde learning

Sheet 80 Mon 14 June 2027 *Year-end review and consolidation*

1. Warm-up · Quick [GEO] $A(-1,4)$ and $B(3,1)$. Coordinates of $\overrightarrow{AB}$?
   Ans. $\begin{pmatrix}4\\-3\end{pmatrix}$.
2. Core · Method [GEO] Equation of (AB) in gradient-intercept form with $A(0,1)$ and $B(2,5)$.
   Ans. Gradient 2; $y = 2x + 1$.
3. Core + · Method [REP] $A(1,0)$, $B(4,1)$, $C(5,4)$. Find D so that $ABCD$ is a parallelogram.
   Ans. $D = A + C - B = (2,3)$; check: $\overrightarrow{AB} = \overrightarrow{DC} = \begin{pmatrix}3\\1\end{pmatrix}$.
4. Review · Quick [CAL] Calculate $-5 + 8 \times (-2)$.
   Ans. $-5 - 16 = -21$.
   Multiplication takes priority.
5. Thread · Statement → formula [REG] The area of a square of side x is a function A of x. Give $A(x)$, a table of values for $x = 1\,;\,2\,;\,3$, and say whether A is proportional to x.
   Ans. $A(x) = x^2$; $1\,;\,4\,;\,9$; not proportional: $A(2) \neq 2A(1)$.

Sheet 81 Tue 15 June 2027 *Year-end review and consolidation*

1. Warm-up · Quick [NUM] Write 0.00045 in standard form.
   Ans. 4.5×10^{-4}.
2. Core · Quick [NUM] Find $[-2,3) \cap (0,5]$.
   Ans. $(0,3)$.
3. Core + · Method [EQ] Solve $|x-1| < 4$ and give S as an interval.
   Ans. $-4 < x - 1 < 4$: $S = (-3,5)$.
4. Review · Expand [DEV] Expand and simplify $(2x-1)^2$.
   Ans. $4x^2 - 4x + 1$.
5. Thread · Quick [GEO] A cylinder has radius 3 cm and height 5 cm. Calculate its exact volume.
   Ans. $V = \pi \times 3^2 \times 5 = 45\pi$ cm^3.

Sheet 82 Thu 17 June 2027 *Year-end review and consolidation*

1. Warm-up · Quick [IMG] $f(x) = x^2 - 4x$. Calculate $f(3)$.
   Ans. $9 - 12 = -3$.
2. Core · Quick [EQ] Solve $x^2 = 9$ and then $|x| = 2$.
   Ans. $x = \pm 3$; $x = \pm 2$.
3. Core + · Method [REG] $f(x) = 2x - 3$ and $g(x) = x + 1$. Solve $f(x) < g(x)$.
   Ans. $2x - 3 < x + 1 \Leftrightarrow x < 4$: $S = (-\infty, 4)$.
4. Review · Quick [QP] A tap delivers 18 L in 4 min. What volume does it deliver in 10 min?
   Ans. $4.5 \times 10 = 45$ L.
5. Thread · Reading code [ALG] What is the final value of `n`?

```
n = 0
while n*n < 30:
    n = n + 1
```

   Ans. $n = 6$ because $5^2 = 25 < 30$ but $6^2 = 36 \geqslant 30$.
   Smallest natural number whose square is greater than or equal to 30.

Sheet 83 Mon 21 June 2027 *Year-end review and consolidation*

1. Warm-up · Quick [STA] Mean of 5, 7, 9, 11?
   Ans. 8.
2. Core · Method [STA] Data set 1 ; 3 ; 4 ; 6 ; 8 ; 9 ; 11: find Q_1, the median and Q_3.
   Ans. $n = 7$: Q_1 has rank $\lceil 7/4 \rceil = 2$: 3; median (rank 4): 6; Q_3 has rank $\lceil 21/4 \rceil = 6$: 9.
3. Core + · True/false [STA] True or false? Justify. Adding 10 to every value of a data set leaves its standard deviation unchanged.
   Ans. True. The deviations from the mean do not change.
4. Review · Reflection [IMG] How many preimages do -1 and 4 have for $f(x) = x^2$?
   Ans. -1: none; 4: two (-2 and 2).
5. Thread · Prove [LIT] Show that for every real x: $(x-1)(x+1)+1 = x^2$.
   Ans. $(x-1)(x+1)+1 = x^2 - 1 + 1 = x^2$.

Sheet 84 Tue 22 June 2027 *Year-end review and consolidation*

1. Warm-up · Quick [DEV] Expand and simplify $(2x-3)(x+4)$.
   Ans. $2x^2 + 8x - 3x - 12 = 2x^2 + 5x - 12$.
2. Core · Quick [EQ] Solve $3x - 7 \leqslant 2x + 5$.
   Ans. $x \leqslant 12$; $S = (-\infty, 12]$.
3. Core + · Method [EQ] Solve $(x-1)(x+4) = 0$ and then $x^2 = 5$.
   Ans. $x = 1$ or $x = -4$; $x = \sqrt{5}$ or $x = -\sqrt{5}$.
4. Review · Reflection [PNP] Membership: 5 € + 3 € per entry. Is the price proportional to the number of entries?
   Ans. No: 1 entry $\mapsto$ 8 €, 2 entries $\mapsto$ 11 $\neq$ 16 €.
5. Thread · Quick [SOL] A cone has radius 3 cm and height 4 cm. Calculate its slant height and its exact volume.
   Ans. Slant height $\sqrt{3^2 + 4^2} = 5$ cm; $V = \frac{1}{3}\pi \times 9 \times 4 = 12\pi$ cm^3.

Sheet 85 Thu 24 June 2027 *Year-end review and consolidation*

1. Warm-up · Quick [PRO] Which multiplier corresponds to an increase of 8%?
   Ans. 1.08.
2. Core · Method [PRO] An increase of 25% is followed by a decrease of 20%. What is the overall rate of change?
   Ans. $1.25 \times 0.8 = 1$: no change (0%).
3. Core + · Quick [PRO] A price goes from 5,000 to 5,750 Ar. Rate of change?
   Ans. $\dfrac{750}{5{,}000} = 15\%$.
4. Review · Recognise [PER] What is the cross-section of a cylinder by a plane containing its axis? By a plane perpendicular to its axis?
   Ans. A rectangle; a disc.
5. Thread · Problem [EQ] Three times a number, minus 4, is equal to twice that number plus 5. Write an equation and then find the number.
   Ans. $3x - 4 = 2x + 5$, so $x = 9$.
   Check: $3 \times 9 - 4 = 23 = 2 \times 9 + 5$.

Sheet 86 Mon 28 June 2027 *Year-end review and consolidation*

1. Warm-up · Quick [MOD] Direction of variation of $x \mapsto -3x + 2$?
   Ans. Strictly decreasing (gradient $-3 < 0$).
2. Core · Method [REG] Minimum and maximum of $f(x) = x^2$ on $[-3, 1]$?
   Ans. Minimum 0 (at 0); maximum 9 (at -3); $f(1) = 1 < 9$.
3. Core + · Method [CMP] For $x > 0$, compare $\dfrac{1}{x}$ and $\dfrac{1}{x+1}$.
   Ans. $0 < x < x + 1$ and the reciprocal function is decreasing on $(0, +\infty)$: $\dfrac{1}{x} > \dfrac{1}{x+1}$.
4. Review · Construction [CON] Describe in three steps the construction of the reflection A' of A in a line (d).
   Ans. 1. Draw the perpendicular to (d) through A, meeting (d) at H. 2. Using compasses, mark off HA on the other side of (d). 3. A' is the point obtained: H is the midpoint of $[AA']$.
5. Thread · Method [DEM] A point I satisfies $IA = IB$. Where does it lie? Which property is used?
   Ans. On the perpendicular bisector of $[AB]$ (characterisation of the perpendicular bisector).

Sheet 87 Tue 29 June 2027 *Year-end review and consolidation*

1. Warm-up · Quick [PRB] $P(A) = 0.28$. Calculate $P(\overline{A})$.
   Ans. 0.72.
2. Core · Method [PRB] 100 people, including 60 women; 6 women and 4 men are left-handed. Calculate $P_F(G)$ and $P_G(F)$.
   Ans. $P_F(G) = \dfrac{6}{60} = 0.1$; $P_G(F) = \dfrac{6}{10} = 0.6$.
3. Core + · Method [PRB] $P(A) = 0.4$, $P_A(B) = 0.5$, $P_{\overline{A}}(B) = 0.2$. Calculate $P(B)$.
   Ans. $0.4 \times 0.5 + 0.6 \times 0.2 = 0.2 + 0.12 = 0.32$.
4. Review · Problem [MOD] Volume of a cube of edge x: $V(x) = x^3$. Calculate $V(3)$ and find x such that $V(x) = 64$.
   Ans. $V(3) = 27$; $x^3 = 64$, $x = 4$.
5. Thread · Quick [REP] In which quadrant (numbered I to IV anticlockwise) does $M(a, b)$ lie, with $a < 0$ and $b > 0$?
   Ans. Quadrant II (top left).

Sheet 88 Thu 1st July 2027 *Year-end review and consolidation*

1. Warm-up · Quick [GEO] Hypotenuse of a right-angled triangle with sides 5 and 12?
   Ans. $\sqrt{25 + 144} = 13$.
2. Core · Quick [REP] Midpoint of $[AB]$ with $A(-3, 2)$ and $B(5, -4)$?
   Ans. $\left(\dfrac{-3+5}{2}; \dfrac{2-4}{2}\right) = (1, -1)$.
3. Core + · Write-up [DEM] Show that $A(0, 0)$, $B(2, 1)$, $C(-1, 2)$ form an isosceles right-angled triangle at A.
   Ans. $AB^2 = 5$, $AC^2 = 5$; $BC^2 = 9 + 1 = 10 = AB^2 + AC^2$: converse of Pythagoras' theorem.
4. Review · Quick [NUM] Give an approximate value of $\dfrac{1}{3}$ correct to 10^{-2}. Is this number a decimal?
   Ans. 0.33. No: the decimal expansion of $\frac{1}{3}$ is unending ($0.333\ldots$).
5. Thread · Problem [DIV] What is the smallest integer that leaves remainder 1 when divided by 4 and by 6, other than 1?

Ans. $1 + \operatorname{lcm}(4, 6) = 13$.